Case 3 · overexpress-tf-q1 (CompBioBench), GPT-5.6 Luna

The confounder was measured and then ignored: AT-rich regions make AT-rich motifs look enriched

Question: We perturbed a primary cell culture with a cocktail of transcription factors (TFs). ATAC-seq is performed on the initial sample and 48 hours after perturbation. Exactly one of the TFs in the following list is part of the cocktail: FOXA1, ASCL1, TFAP2A, RUNX1, KLF4, PAX7, TCF7, REST, SPI1, GATA3. Identify which one.

